Testing one-hotness of a multilinear polynomial over a binary field

- $\mathbb{F}$ is a cryptographically large binary field ($\mathbb{F}_{2^{128}}$, $\mathbb{F}_{2^{192}}$, ...).
- Bold letters are tuples: $\mathbf{X} = (X_1, \ldots, X_m)$, $\mathbf{1} + \mathbf{X} = (1 + X_1, \ldots, 1 + X_m)$, $\mathbf{Y} = (Y_1, \ldots, Y_n)$.
- $\mathbb{F}^{<2}[\mathbf{X}]$ is the set of multilinear polynomials in $X_1, \ldots, X_m$.
- $f \in \mathbb{F}^{<2}[\mathbf{X}]$ is *one-hot* if $f = \mathrm{eq}(\mathbf{X}, \mathbf{u})$ for some $\mathbf{u} \in \{0,1\}^m$: on the cube, it is 1 at $\mathbf{u}$ and 0 elsewhere.

Given a committed $A \in \mathbb{F}^{<2}[\mathbf{X}, \mathbf{Y}]$, we want to check that $A(\cdot, \mathbf{j})$ is one-hot for every $\mathbf{j} \in \{0,1\}^n$.

The trick. For any $f \in \mathbb{F}^{<2}[\mathbf{X}]$:

$$\boxed{f \text{ is one-hot} \iff f(\mathbf{X}) \cdot f(\mathbf{1} + \mathbf{X}) = \prod_{h=1}^{m} X_h(1 + X_h)} \tag{1}$$

Proof. ($\Rightarrow$) Let $f = \mathrm{eq}(\mathbf{X}, \mathbf{u})$. It has one factor per coordinate: X_h if $u_h = 1$, and $1 + X_h$ if $u_h = 0$. In $f(\mathbf{1} + \mathbf{X})$ each factor becomes the other one. So coordinate h contributes $X_h(1 + X_h)$ to $f(\mathbf{X}) \cdot f(\mathbf{1} + \mathbf{X})$, whatever u_h is.

($\Leftarrow$) Write $\mathbf{X}_{<m} = (X_1, \ldots, X_{m-1})$, so that $f(\mathbf{X}) = (1 + X_m) \cdot f(\mathbf{X}_{<m}, 0) + X_m \cdot f(\mathbf{X}_{<m}, 1)$.

Set $X_m = 0$ in (1). The right side vanishes, so $f(\mathbf{X}_{<m}, 0) \cdot f(\mathbf{1} + \mathbf{X}_{<m}, 1) = 0$, and one of $f(\mathbf{X}_{<m}, 0)$, $f(\mathbf{X}_{<m}, 1)$ is the zero polynomial. Hence $f = X_m \cdot g$ or $f = (1 + X_m) \cdot g$, for some $g \in \mathbb{F}^{<2}[\mathbf{X}_{<m}]$. In both cases

$$f(\mathbf{X}) \cdot f(\mathbf{1} + \mathbf{X}) = X_m(1 + X_m) \cdot g(\mathbf{X}_{<m}) \cdot g(\mathbf{1} + \mathbf{X}_{<m}),$$

so, dividing by $X_m(1 + X_m)$, g satisfies (1) with one variable less. Repeating for each variable, f is a constant c times one factor per coordinate, X_h or $1 + X_h$, and $c^2 = 1$. So $c = 1$, because $c^2 + 1 = (c+1)^2$ (the only part where characteristic 2 matters). Hence $f = \mathrm{eq}(\mathbf{X}, \mathbf{u})$, where $u_h = 1$ if the factor is X_h and $u_h = 0$ if it is $1 + X_h$. □

Protocol. Every $A(\cdot, \mathbf{j})$ is one-hot if and only if, as polynomials in $\mathbf{X}$ and $\mathbf{Y}$,

$$\sum_{\mathbf{j} \in \{0,1\}^n} \mathrm{eq}(\mathbf{Y}, \mathbf{j}) \cdot A(\mathbf{X}, \mathbf{j}) \cdot A(\mathbf{1} + \mathbf{X}, \mathbf{j}) = \prod_{h=1}^{m} X_h(1 + X_h). \tag{2}$$

After A is committed, the verifier samples random $\mathbf{r} \in \mathbb{F}^m$ and $\boldsymbol{\rho} \in \mathbb{F}^n$, and (2) is checked at $(\mathbf{X}, \mathbf{Y}) = (\mathbf{r}, \boldsymbol{\rho})$ via a degree-3 sumcheck.